LittleStory iOS LLD

LittleStory iOS — Low-level design

Status: Draft v0.2 for owner review · 11 Nov 2026 (v0.2: iPad adaptive layout §9.4, exact latest versions, minimum iOS 26) **Companion:** High-level design. Read the HLD first; this document assumes its decisions (native Swift 6.4 + SwiftUI, Xcode 27.2, iPhone + iPad universal app, minimum iOS 26, one local package, no paid services, mocked AI). **Reference:** Android LLD (`development` @ `5c505c6`). Section numbers in brackets, such as [A-4.3], point to the matching Android LLD section.

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1. Package and targets

1.1 Folder layout

```
Packages/LittleStoryKit/
├── Package.swift
├── Sources/
│   ├── Common/            AppConfig.swift, FeatureFlags.swift,
SupportedLanguage.swift, AppInfo.swift, ParentalGate.swift
│   ├── Domain/            Models/, Validation/, Repositories/, Services/,
Simulation/
│   ├── Data/              Content/, Persistence/, Preferences/, Repositories/,
Simulated/, Resources/Mock/*.json
│   ├── DesignSystem/      Tokens.generated.swift, Theme.swift, Fonts/, Icons/,
Components/
│   ├── CoreUI/            Chrome/, Story/, Illustration/, Localization/,
Safety/, Loading/
│   ├── FeatureOnboarding/ Onboarding/, Setup/, OnboardingRoute.swift,
Localizable.xcstrings
│   ├── FeatureHome/ …     (same shape for Discover, Create, Reader, Library,
Settings)
│   └── TestSupport/       Fakes/, Fixtures/ (clocks come from swift-clocks)
└── Tests/
    ├── CommonTests/  DomainTests/  DataTests/  CoreUITests/
    └── Feature*Tests/     ViewModel tests + snapshot tests (__Snapshots__/)
```

1.2 `Package.swift` (shape)

```
        // swift-tools-version: 6.4
        import PackageDescription

        let package = Package(
            name: "LittleStoryKit",
            defaultLocalization: "en",
            platforms: [.iOS(.v26)],
            products: [ .library(name: "LittleStoryKit", targets:
["AppFeatures", "Data"]) ],
            dependencies: [
                .package(url: "https://github.com/apple/swift-async-
algorithms", exact: "1.1.7"),
                .package(url: "https://github.com/pointfreeco/swift-clocks",
exact: "1.1.1"),
                .package(url: "https://github.com/pointfreeco/swift-snapshot-
testing", exact: "1.19.7"),
            ],
            targets: [
                .target(name: "Common"),
                .target(name: "Domain", dependencies: ["Common",
                        .product(name: "AsyncAlgorithms", package: "swift-
async-algorithms")]),
                .target(name: "Data", dependencies: ["Domain", "Common"],
resources: [.process("Resources")]),
                .target(name: "DesignSystem", resources: [.process("Fonts"),
.process("Icons")]),
                .target(name: "CoreUI", dependencies: ["DesignSystem",
"Domain"], resources: [.process("Localizable.xcstrings")]),
                .target(name: "FeatureHome", dependencies: ["CoreUI",
"DesignSystem", "Domain"]),
                // … one target per feature, identical dependency list
                .target(name: "AppFeatures", dependencies:
["FeatureOnboarding", "FeatureHome", "FeatureDiscover",
                        "FeatureCreate", "FeatureReader", "FeatureLibrary",
"FeatureSettings"]),
                .target(name: "TestSupport", dependencies: ["Domain"]),
                .testTarget(name: "DomainTests", dependencies: ["Domain",
"TestSupport",
                        .product(name: "Clocks", package: "swift-clocks")]),
                .testTarget(name: "FeatureHomeTests", dependencies:
["FeatureHome", "TestSupport",
                        .product(name: "SnapshotTesting", package: "swift-
snapshot-testing")]),
            ],
            swiftLanguageModes: [.v6]
        )
```

- Versions are the newest stable on 11 Nov 2026 and pinned with `exact:` plus `Package.resolved`; upgrades come as one-dependency PRs.
- Every target turns on `StrictConcurrency` (default in Swift 6 mode) and `-warnings-as-errors` in CI.

1.3 Allowed imports (checked by `Tools/check-dependencies`)

Target	May import
`Common`	Foundation
`Domain`	`Common`, Foundation, AsyncAlgorithms
`Data`	`Domain`, `Common`, Foundation, SwiftData, OSLog
`DesignSystem`	SwiftUI
`CoreUI`	`DesignSystem`, `Domain`, `Common`, SwiftUI
`Feature*`	`CoreUI`, `DesignSystem`, `Domain`, `Common`, SwiftUI
App target	everything

The script parses `import` lines per source folder and fails CI on any edge not in this table (the iOS twin of Android `DependencyRules.kt`).

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2. `Common` [A-2]

Type	Kind	Holds
`AppConfig`	`enum` namespaces of `static let`	all limits, options and timings (below)
`FeatureFlags`	`struct`, `Sendable`	`demoSubscription`, `draftLegalText`, `onDeviceNarration`, `accounts` (all default off except demo)
`SupportedLanguage`	`struct` + registry	`tag`, `nativeName`, `isEnabled`, `isRightToLeft`
`AppInfo`	`struct`	version, build, from `Bundle.main` (injected)
`ParentalGateChallenge`	`struct`	`a × b`, `isAnswered(by:)`; `ParentalGateChallenges` protocol with random and fixed versions

```
        public enum AppConfig {
            public enum Create {
                public static let promptMaxCharacters = 500
                public static let durationOptionsMinutes = [5, 8, 10, 15]
                public static let defaultDurationMinutes = 8
                public static let themes = ["adventure", "bedtime", "animals",
"friendship", "fantasy", "learning", "indian_folklore"]
                public static let moods = ["calm", "funny", "magical",
"adventurous"]
            }
            public enum Profile {
                public static let nameMaxCharacters = 30
                public static let ageRange = 3...12
                public static let defaultAge = 6
                public static let durationOptionsMinutes = [5, 10, 15]   //
differs from Create, as on Android (open question C1)
                public static let interestIDs = ["animals", "cricket",
"adventure", "space", "nature", "music"]
                public static let guestKey = "guest"
            }
            public enum Generation {
                public static let stepPercent = 4
                public static let stepInterval: Duration = .milliseconds(120)
                public static let restartPercent = 8
                public static let stageThresholdsPercent = (imagining: 38,
writing: 76)
                public static let readyDelay: Duration = .milliseconds(500)
            }
            public enum Playback {
                public static let tick: Duration = .milliseconds(500)
                public static let tickFraction = 0.0025
                public static let skipSeconds = 15
                public static let speeds: [Double] = [0.75, 1.0, 1.25, 1.5]
                public static let saveEveryTicks = 10
                public static let sleepTimerMinutes = [10, 15, 30]
            }
            public enum Reader { public static let textScaleRange = 0.9...1.3;
public static let textScaleStep = 0.1 }
            public enum Downloads { public static let simulatedStoryBytes:
Int64 = 12_000_000; public static let simulatedQuotaBytes: Int64 =
1_000_000_000 }
            public enum UI { public static let loadingIndicatorDelay: Duration
= .milliseconds(300) }
        }

        public struct SupportedLanguage: Sendable, Hashable, Identifiable {
            public let tag: String          // BCP-47: "en", "kn"
            public let nativeName: String   // the language's own name; not
translated by design
            public let isEnabled: Bool
            public var id: String { tag }

            public static let english = SupportedLanguage(tag: "en",
nativeName: "English", isEnabled: true)
            public static let kannada = SupportedLanguage(tag: "kn",
nativeName: "ಕನ್ನಡ", isEnabled: true)
            public static let all: [SupportedLanguage] = [.english, .kannada]
```

```
            public static let fallback = english
            public static var enabled: [SupportedLanguage] {
all.filter(\.isEnabled) }
        }
```

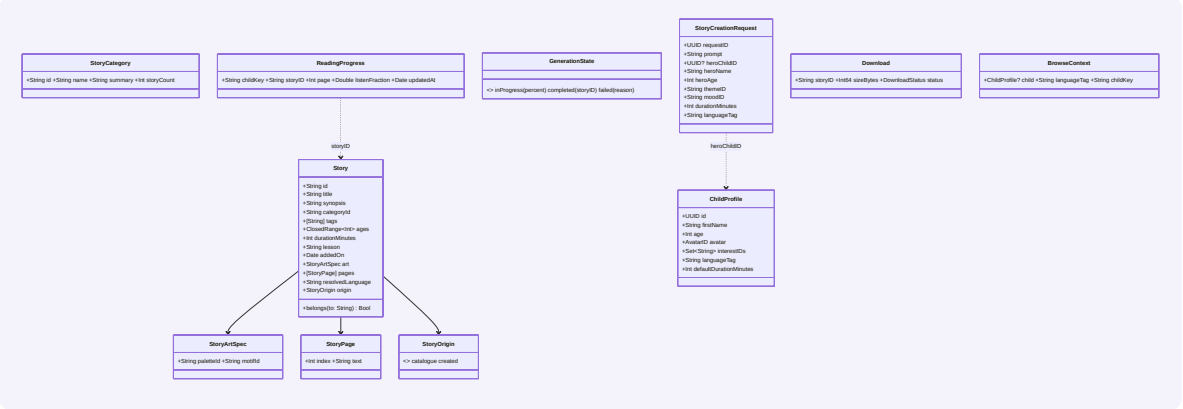
Values copy the Android `AppConfig` (`core/common/.../config/AppConfig.kt` on `development`) so both apps behave the same; the snippet shows a subset, the full list is ported in ticket LS-iOS-010. The Android debug-only Gemini prototype adds `writerWaitPercent = 92` and `writerTimeout = 90 s` for a real writer; iOS keeps the same two values ready for a future `StoryWriter`.

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3. `Domain` [A-4]

Pure Swift. Every type is a `struct` or `enum`, `Sendable`, `Hashable` where useful. No localized text: content strings (story titles) arrive already resolved for a language; UI copy lives in feature string catalogs.

3.1 Models



Other models (same meaning as Android): `SavedStory`, `PlaybackState`, `SleepTimer` (`.off`, `.minutes(Int)`, `.endOfStory`), `ReadingPreferences(textScale, playbackSpeed)`, `ReminderSettings(enabled, time, weekdays)`, `ParentSettings`, `NotificationItem` (+ `section`, `kind`), `Plan` (price in minor units + ISO currency), `ChildProfileDraft`.

- `childKey` is the child's id, or the constant guest key when no profile exists (Skip in Setup).
- `Story` is a value type; a created story uses id `generated_<requestID>` so a retry never duplicates.

3.2 Validation [A-4.2]

Validator	Input	Errors
`PromptValidator.validate(_:)`	prompt	`.empty` (blank after trimming), `.tooLong` (> 500)

Validator	Input	Errors
`ProfileValidator.validate(_:)`	`ChildProfileDraft`	`Set<ProfileError>`: `.nameEmpty`, `.nameTooLong` (> 30), `.ageOutOfRange` (not 3…12), `.durationNotAllowed`

Validators are pure `static func`s returning `Result` or a `Set`, so they are trivially unit tested.

3.3 Repository and service protocols [A-4.3]

Reads are `AsyncStream`s that emit the current value at once and again on every change. Writes are `async`. All protocols are `Sendable`.

Protocol	Reads	Writes
`CatalogRepository`	`categories(language:)`, `stories(language:)`, `story(id:language:)`	—
`ProfilesRepository`	`profiles()`, `activeProfile()`	`create(_:)`, `update(_:)`, `delete(id:)`, `setActive(`
`FavouritesRepository`	`favouriteIDs(childKey:)`	`setFavourite(_:storyID:` `toggle(storyID:childKey`
`LibraryRepository`	`created(childKey:)`, `story(id:)`	`save(_:childKey:)`
`GenerationRepository`	`request(id:)`, `generate(id:startPercent:) -> AsyncStream<GenerationState>`	`submit(_:)`
`ProgressRepository`	`progress(childKey:)`	`update(_:)`
`DownloadsRepository`	`downloads()`	`start(storyID:)`, `remove`
`PreferencesRepository`	`firstRunCompleted()`, `contentLanguage()`, `reading()`, `parentSettings()`	setters
`NotificationsRepository`	`notifications(language:)`	`markRead(id:)`, `markAllR`
`SubscriptionRepository`	`plans()`	—
`LocalDataRepository`	—	`deleteAll()`
AI services		
`StoryWriter`	—	`write(_ request: StoryCreationRequest) as -> StoryDraft`
`Narrator`	`events() -> AsyncStream<NarrationEvent>`	`load(_:)`, `play()`, `pause` `seek(to:)`, `setRate(_:)`

Protocol	Reads	Writes
`Illustrator`	—	`art(for: StoryDraft) asy` `StoryArtSpec`

```
        public protocol CatalogRepository: Sendable {
            func categories(language: String) -> AsyncStream<[StoryCategory]>
            func stories(language: String) -> AsyncStream<[Story]>
            func story(id: String, language: String) -> AsyncStream<Story?>
        }

        public protocol StoryWriter: Sendable {
            /// MVP: TemplateStoryWriter. Later: on-device or remote writers.
Must never log the prompt.
            func write(_ request: StoryCreationRequest) async
throws(StoryWriterError) -> StoryDraft
        }
```

`StoryStores` bundles catalogue, library, favourites, downloads and progress for Reader, Player and My Stories, exactly as on Android.

3.4 Domain services (instead of thin use cases)

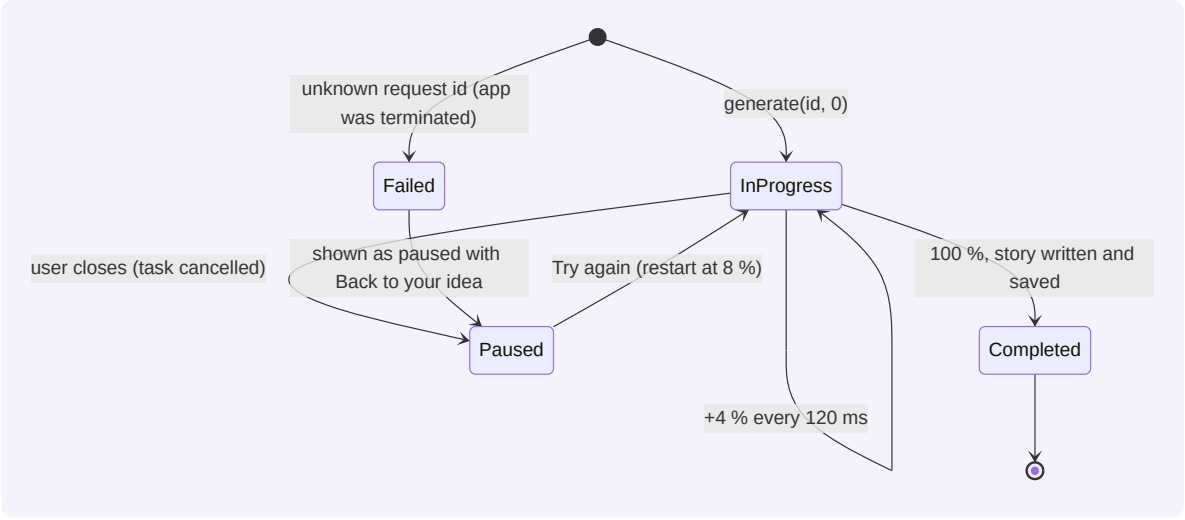
Only logic that is shared and non-trivial becomes a type; there is no use case that only forwards a call.

Service	Job
`BrowseContextProvider`	combines `activeProfile()` and `contentLanguage()` into `AsyncStream<BrowseContext>`
`StoryLookup`	catalogue story, else created story, for a context (`observeStory` on Android)
`SearchStories`	locale-aware, diacritic-insensitive search over title, synopsis and category name
`GenerationEngine`	runs `GenerationProgress` then calls `StoryWriter` and `LibraryRepository.save`

3.5 Simulation rules [A-4.4]

Generation progress — `GenerationProgress.run(from:clock:)` yields `.inProgress(p)` immediately, then `+4` every 120 ms on the injected clock until 100. Cancelling the consuming task pauses it.

```
        public struct GenerationProgress: Sendable {
            let clock: any Clock<Duration>
            public func run(from start: Int) -> AsyncStream<Int> {
                AsyncStream { continuation in
                    let task = Task {
                        var percent = start
                        continuation.yield(percent)
                        while percent < 100, !Task.isCancelled {
                            try? await clock.sleep(for:
AppConfig.Generation.stepInterval)
                            percent = min(percent +
AppConfig.Generation.stepPercent, 100)
                            continuation.yield(percent)
                        }
                        continuation.finish()
                    }
                    continuation.onTermination = { _ in task.cancel() }
                }
            }
            public static func stage(for percent: Int) -> GenerationStage { /*
imagining < 38 ≤ writing < 76 ≤ preparingNarration */ }
        }
```



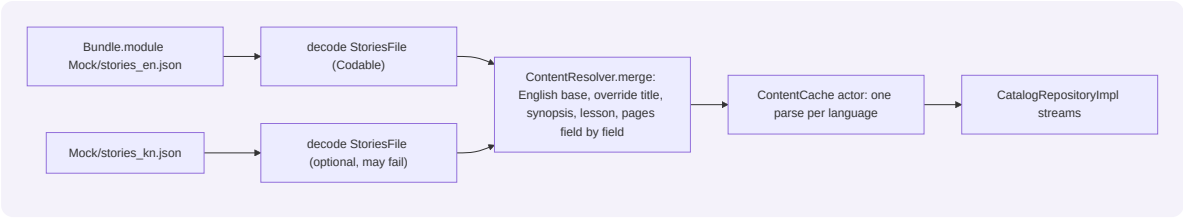
Playback — `SimulatedPlayback` pure functions on a position fraction 0…1:

Function	Rule
`totalSeconds(minutes:)`	`max(minutes, 1) × 60`
`tick(_:speed:)`	`+ 0.0025 × speed`, clamped to 1
`skip(_:minutes:direction:)`	`± 15 / totalSeconds`
`chapter(of:pageCount:)`, `chapterStart(_:pageCount:)`	pages spread evenly
`nextSpeed(after:)`	cycles 0.75 → 1.0 → 1.25 → 1.5
`sleepAfterSeconds(_:)`	`.minutes(n)` → `n × 60`, else `nil`

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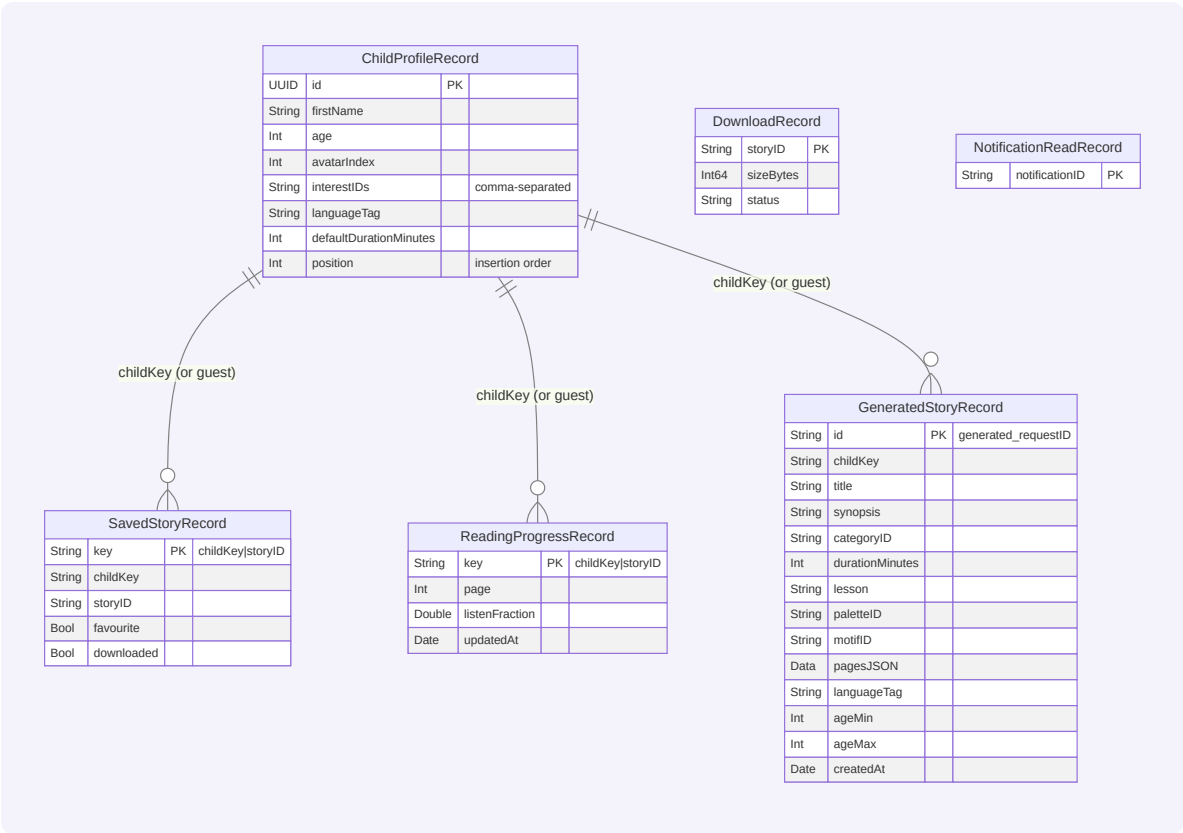
4. `Data` [A-5]

4.1 Content loading and the English fallback



- `MockContentSource` protocol: `func data(named: String) -> Data?`. The app reads `Bundle.module`; tests read fixtures.
- Decoding uses `JSONDecoder` with lenient structs (unknown keys ignored by default in `Codable`; optional fields for anything a translation may omit).
- A blank or missing translated field keeps English; `resolvedLanguage` is the target language only when the title was translated.
- `ContentCache` is an `actor` holding `[String: ResolvedCatalogue]`; concurrent first reads await one parse task.

4.2 SwiftData schema

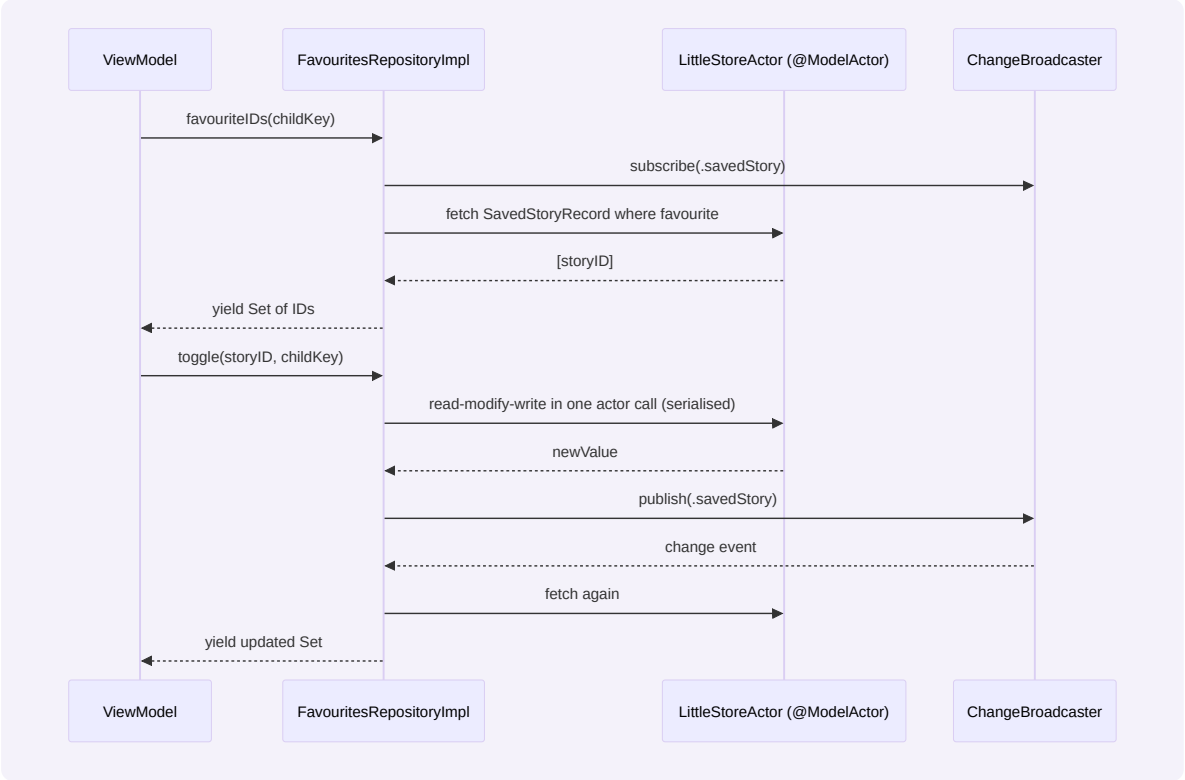


- `@Model` classes use `@Attribute(.unique)` on the key; composite keys are a joined string because SwiftData has no composite primary key.
- No relationships, like Android: the guest key has no profile row. `ProfilesRepositoryImpl.delete` removes that child's rows explicitly in one transaction.

- Schema is versioned from day one: `LittleStorySchemaV1: VersionedSchema` and `LittleStoryMigrationPlan: SchemaMigrationPlan`.
- `@Model` objects never leave the Data layer: each record maps to a Domain struct (`record.toDomain()`, `Record(domain)`), because model objects are not `Sendable`.

4.3 Store access and change streams

SwiftData only auto-observes through `@Query` inside views, which would leak storage into the UI. So the Data layer provides its own observation:



- `LittleStoreActor` is a `@ModelActor`, so all reads and writes are serialised off the main thread; this replaces the Android `Mutex` on favourites.
- `ChangeBroadcaster` is an `actor` that fans out `AsyncStream<Table>` events to subscribers; each repository re-queries only for its table.

4.4 Fallback: GRDB

If SwiftData blocks us (migration bug, performance, concurrency warnings), swap the store for **GRDB.swift** (MIT): SQL schema mirroring Android Room, `DatabaseMigrator`, and `ValueObservation` for streams. Only `Data/Persistence/` changes; repository protocols and tests stay. Decision record: ADR-iOS-004.

4.5 Preferences

`PreferencesStore` is an `actor` over `UserDefaults(suiteName:)` that also broadcasts changes. Keys match the Android DataStore keys.

Key	Type	Default
`first_run_completed`	Bool	false
`content_language_tag`	String	`en`
`ui_language_tag`	String?	nil (follow system)
`active_profile_id`	String?	nil
`reader_text_scale`	Double	1.0
`playback_speed`	Double	1.0
`reminder_enabled`, `reminder_hour`, `reminder_minute`, `reminder_days`	Bool, Int, Int, [String]	false, 20, 30, all days
`notifications_enabled`, `wifi_only_downloads`	Bool	true, true

4.6 Repository implementations

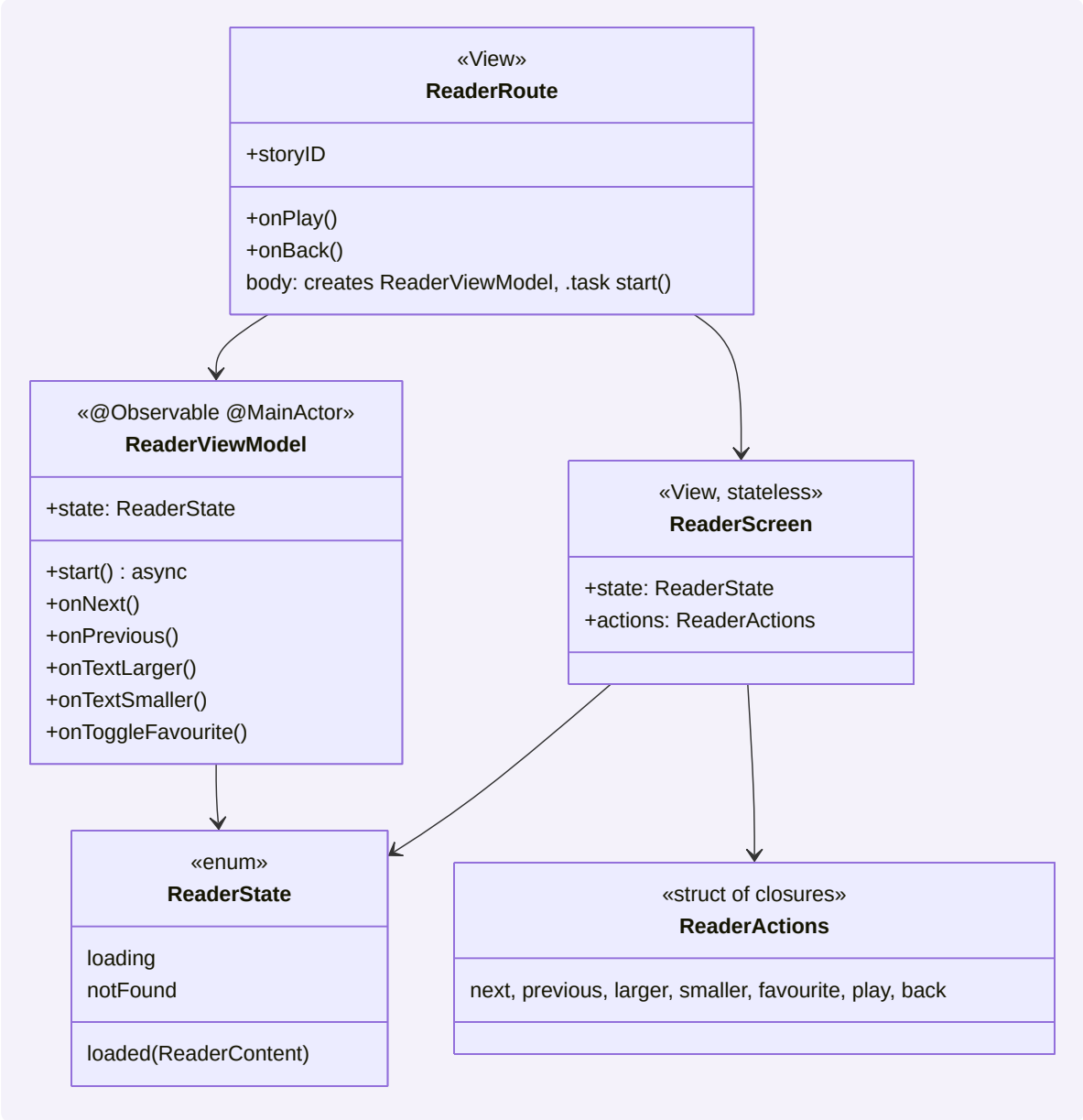
Implementation	Backing	Key behaviour
`CatalogRepositoryImpl`	`MockContentSource` + `ContentCache`	§4.1; `storyCount` comput from `belongs(to:)`
`GenerationRepositoryImpl`	in-memory `[UUID: StoryCreationRequest]` in an actor + `GenerationEngine`	`generate` emits progress, then writes and saves the story, then `.completed`; unknown id → `.failed(.unknownReques` prompt never stored
`TemplateStoryWriter` (mock AI)	`generated_{lang}.json`	fills `{child}` with the hero first name or the template's default hero; theme → category; ages around the hero's age
`LibraryRepositoryImpl`	`GeneratedStoryRecord`	pages encoded as JSON array; unreadable → no pag → Reader Not Found
`ProfilesRepositoryImpl`	`ChildProfileRecord` + preferences	trims names, appends at `max(position) + 1`, first profile becomes active; `activeProfile` falls back the first profile
`FavouritesRepositoryImpl`	`SavedStoryRecord`	toggle serialised in the stor actor; keeps `downloaded`
`ProgressRepositoryImpl`	`ReadingProgressRecord`	upsert per child and story; Reader and Player share or row
`DownloadsRepositoryImpl`	`DownloadRecord`	`start` writes a completed record with the simulated s

Implementation	Backing	Key behaviour
`NotificationsRepositoryImpl`	`notifications_{lang}.json` + read records	merges read ids
`SubscriptionRepositoryImpl`	`plans.json`	demo only
`PreferencesRepositoryImpl`	`PreferencesStore`	§4.5
`LocalDataRepositoryImpl`	store + preferences	deletes all records in one transaction, then clears the defaults suite, then broadca every table

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5. Presentation pattern [A-3]

5.1 Anatomy of a screen



5.2 View model template

```
        @MainActor @Observable
        public final class ReaderViewModel {
            public private(set) var state: ReaderState = .loading

            private let storyID: String
            private let stores: StoryStores
            private let browse: BrowseContextProvider
            private let preferences: PreferencesRepository
            private var page = 0

            public init(storyID: String, stores: StoryStores, browse:
BrowseContextProvider, preferences: PreferencesRepository) { … }

            /// Called from `.task {}`; cancelled automatically when the view
disappears.
            public func start() async {
                for await context in browse.contexts() {
                    for await content in combineLatest(stores.lookup.story(id:
storyID, context: context),
                                                       preferences.reading(),

stores.favourites.favouriteIDs(childKey: context.childKey)) {
                        state = ReaderState(content, page: page)
                    }
                }
            }

            public func onNext() { move(by: 1) }
            public func onPrevious() { move(by: -1) }
        }
```

Rules:

1. `state` is the only stored property the view reads; it is `private(set)`.
2. One public method per user action; no `send(Action)` switch needed.
3. Long-lived observation happens in `start()`, called from `.task`, so SwiftUI cancels it when the screen leaves.
4. Writes are fire-and-forget `Task { await repo.write() }` owned by the view model; the store's stream brings the new value back (single source of truth).
5. Text is `LocalizedStringResource` or an enum; view models never format user-visible strings themselves.
6. Loading spinners appear only after `AppConfig.UI.loadingIndicatorDelay` to avoid flashes.

5.3 One-off effects

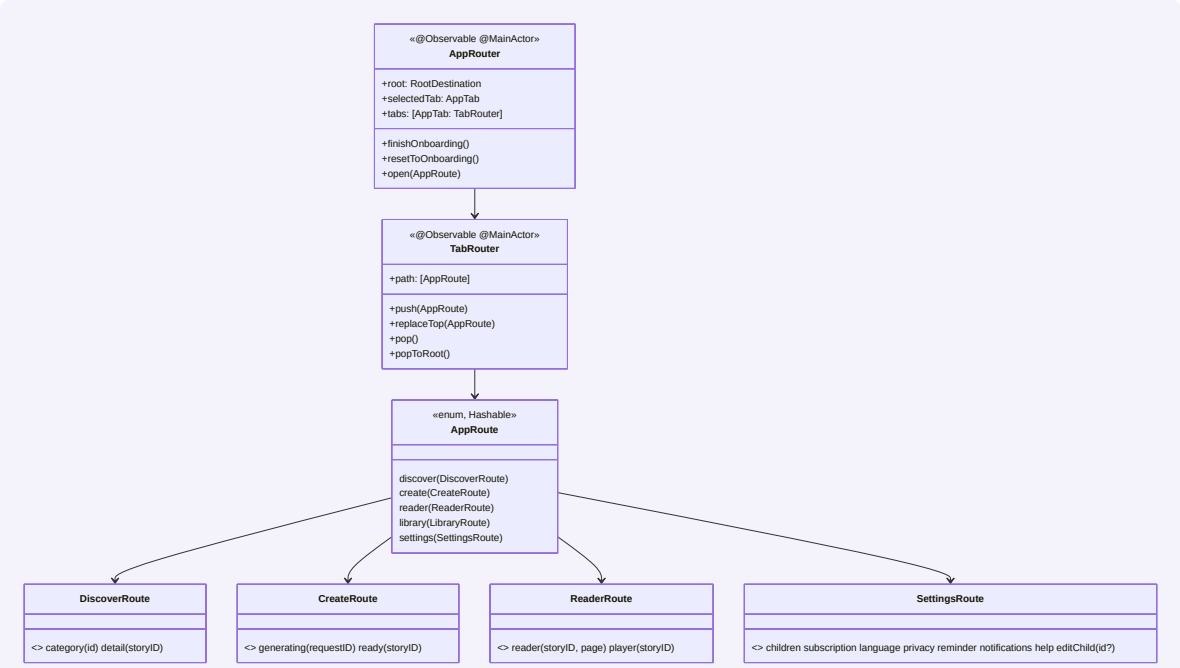
Effect	Example	Delivery
Navigate after work	Generation finished → Story Ready	`AsyncStream<GeneratingEffect>` read by `GeneratingRoute`

Effect	Example	Delivery
Apply something outside the screen	Setup picks UI language	`SetupEffect.applyLanguage(tag)` → `Localizer`
Reset the app	Data deleted	`PrivacyEffect.deleted` → `AppRouter.resetToOnboarding()`

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6. Navigation [A-5 HLD]

6.1 Types



- Each feature exports its own route enum and a `destination(for:)` view builder, plus a `Navigation` struct of closures it needs (`openStory`, `openPlayer`, `back`).
- The app target maps feature closures to router calls, so features never see the router or each other (the Android `AppNavHost` role).
- `RootView` switches on `router.root`: `.onboarding` (Onboarding → Setup in its own stack) or `.main` (TabView).

6.2 Rules

Rule	Implementation
No double opens on fast taps	`TabRouter.push` ignores a route equal to the current top and taps while a transition is in flight
Reader ↔ Player swap	`replaceTop(.reader(.player(id)))`
Story Ready back goes to Create root	on `.ready`, path becomes `[.create(.ready(id))]` with Generating removed

Rule	Implementation
Generating back pauses first	custom back button: first tap pauses, second leaves
Parent-only screens	destination wrapped in `ParentGated { … }` (§9.3)
Tab re-select	tapping the active tab calls `popToRoot()`

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7. Dependency injection and app start [A-6]

7.1 Composition root

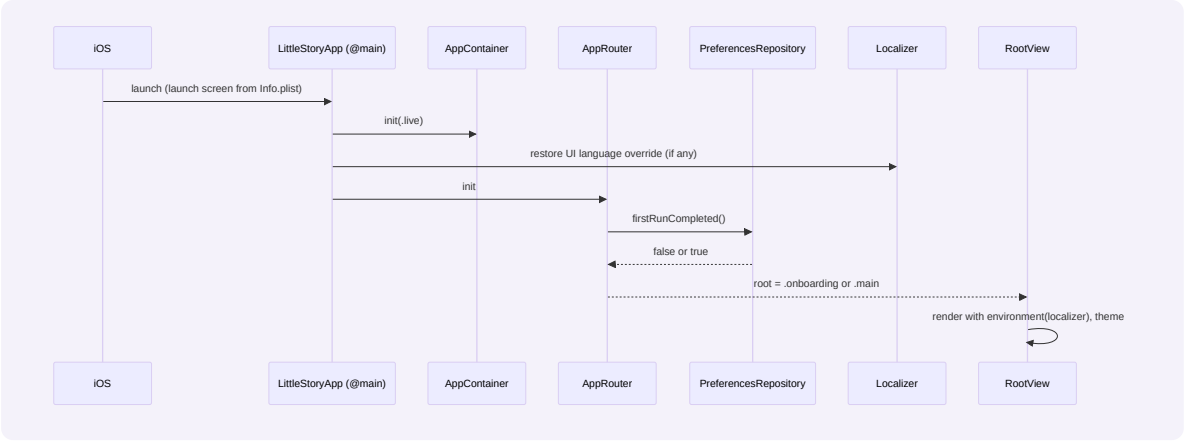
```
        @MainActor
        final class AppContainer {
            let flags: FeatureFlags
            let preferences: PreferencesRepository
            let catalog: CatalogRepository
            let profiles: ProfilesRepository
            let generation: GenerationRepository
            // … every protocol, created once

            init(environment: AppEnvironment = .live) {
                flags = environment.flags
                let store = try! LittleStore.make(inMemory:
environment.inMemoryStore) // only crash point, at launch, by design
                let content = BundleContentSource(bundle: .dataModule)
                preferences = PreferencesRepositoryImpl(store:
PreferencesStore(defaults: environment.defaults))
                catalog = CatalogRepositoryImpl(source: content)
                let writer: StoryWriter = TemplateStoryWriter(source: content)
// mock AI
                generation = GenerationRepositoryImpl(writer: writer, library:
…, clock: environment.clock)
                …
            }

            func makeReaderViewModel(storyID: String) -> ReaderViewModel { … }
// factories per screen
        }
```

- `AppEnvironment` holds `clock`, `defaults`, `flags`, `inMemoryStore`, `gateChallenges`; `.live`, `.preview` and `.uiTest` variants exist.
- UI tests launch with `-uiTest` and get the fixed parental gate, an in-memory store and an immediate clock.
- `ViewModelFactory` protocols per feature are passed to feature roots, so features never see `AppContainer`.

7.2 Startup

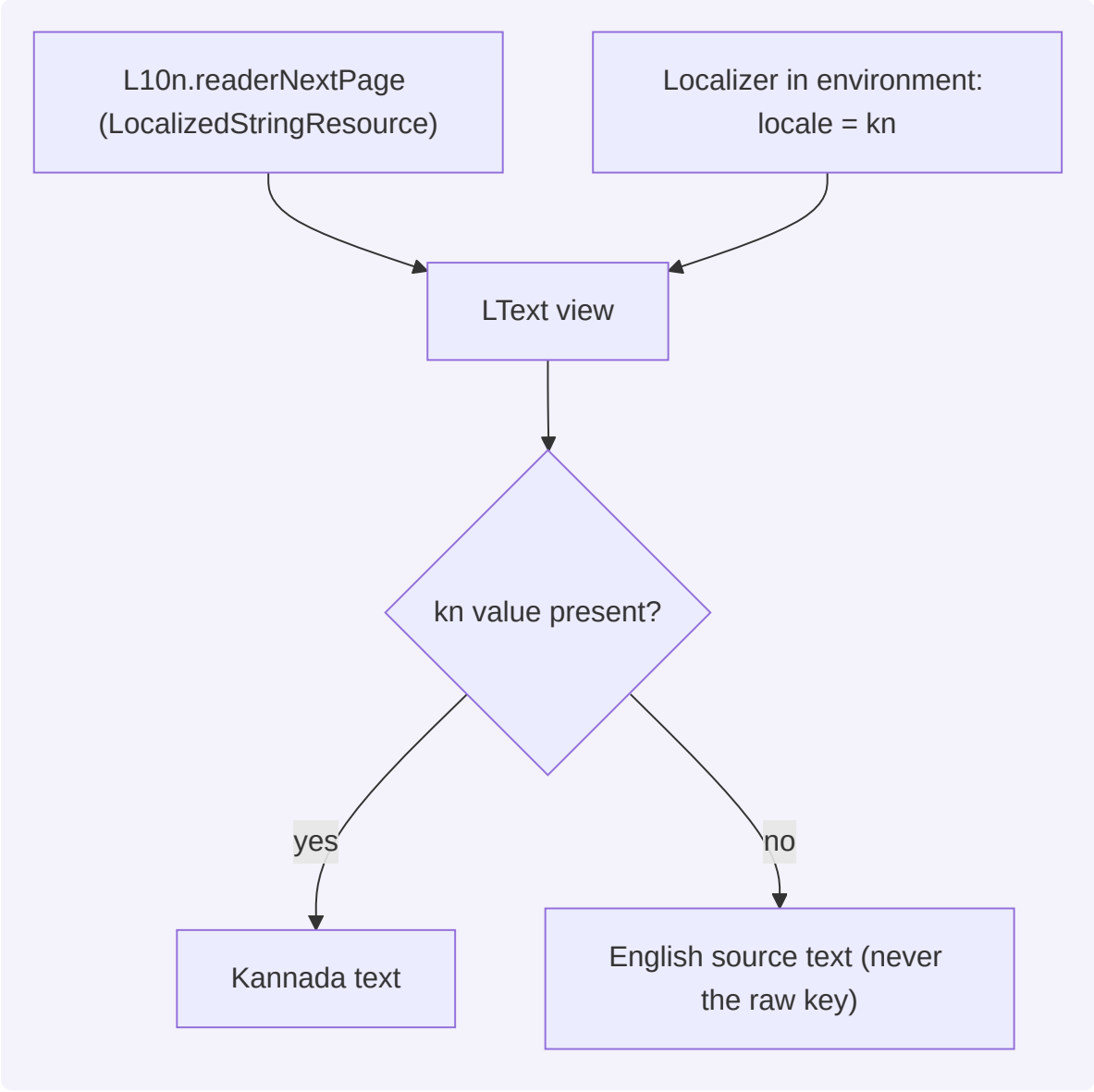


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8. Localization [A-7.6]

8.1 Pieces

Piece	Where	Job
`Localizable.xcstrings`	each UI target	keys, English sou variants, commer
`L10n` keys	generated per target (`Tools/gen-l10n`)	`static let rea` `LocalizedStrin` `bundle: .module` views
`Localizer`	`CoreUI`	`@Observable` ho resolves a resour using that locale injected into the S
`LText`	`CoreUI`	`Text` wrapper: reads the `Locali`
`\.locale`	root view	set from `Localiz` `FormatStyle` fo
Content language	`PreferencesRepository.contentLanguage()`	drives every cata



8.2 Rules

1. Keys are dotted lower snake case, `feature.screen.element`, and equal the Android string names where one exists (`home_title` ↔ `home.title` via the import tool's mapping table).
2. Arguments are positional (`%1$@`, `%2$lld`); plurals use String Catalog variants (Kannada uses `other` plus `one`).
3. Accessibility labels, hints and values are catalog strings too.
4. Native language names come from `SupportedLanguage.nativeName`, never translated.
5. New Kannada strings are marked `needs_review` in the catalog and **Draft — needs native review** in the workbook.

8.3 Adding a language (example: Hindi)

1. Add `SupportedLanguage(tag: "hi", nativeName: "हिन्दी", isEnabled: true)`.
2. Add `hi` to every `.xcstrings` and to `CFBundleLocalizations` in Info.plist.

3. Add `stories_hi.json`, `categories_hi.json`, `generated_hi.json`, `notifications_hi.json` (copied from Android when they exist).
4. Add a Devanagari font if the system font is not enough.
5. Add `hi` to the snapshot matrix. No feature code changes.

---

9. Design system, `CoreUI` and adaptive layout [A-10]

9.1 Tokens

`Tools/gen-tokens` reads `docs/design/design-tokens.json` (copied from the Android repo) and writes `Tokens.generated.swift`:

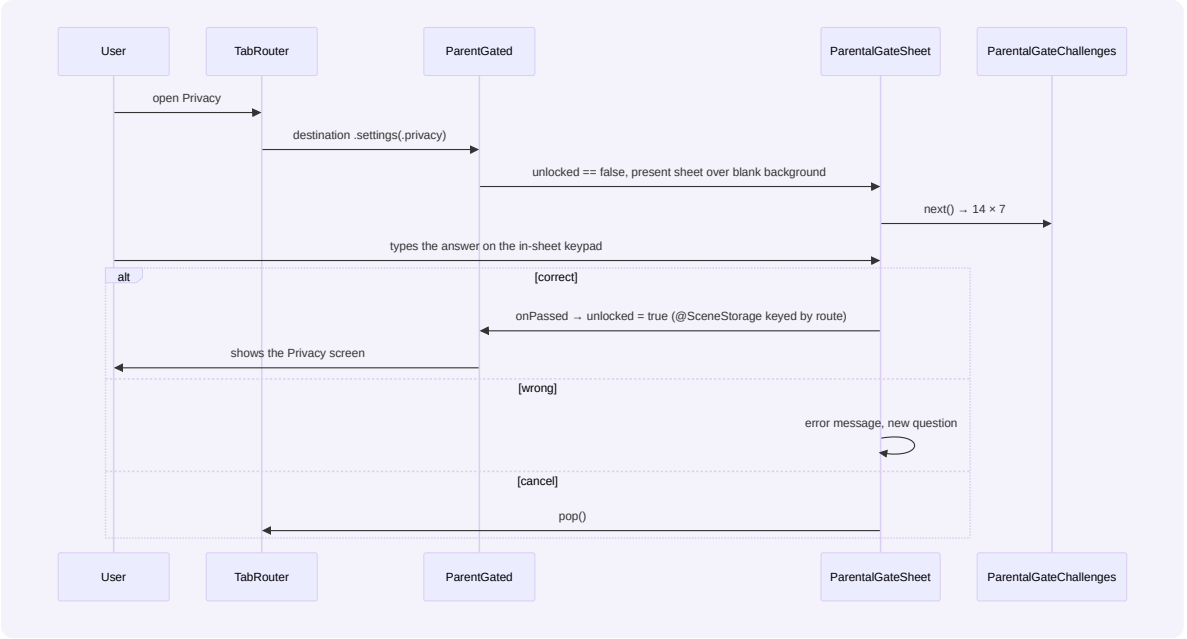
```
        public enum LSColor { public static let primary = Color(lsHex:
0x6C4CF1) /* generated */ … }
        public enum LSSpacing { public static let xs: CGFloat = 4, sm: CGFloat
= 8, md: CGFloat = 16, lg: CGFloat = 24 }
        public enum LSRadius { public static let card: CGFloat = 20, chip:
CGFloat = 999 }
        public enum LSMotion { public static let standard: Animation =
.easeInOut(duration: 0.25) }
```

- Typography uses `Font.custom(_:size:relativeTo:)` so every style scales with Dynamic Type; Kannada styles add line spacing.
- `LSTheme` is injected with `@Entry` environment values; night surfaces for Reader and Player.
- Hex values above are placeholders; real values come from the generated file.

9.2 Components

Component	Target	Notes
`LSButton` (primary, secondary, text)	DesignSystem	min 44 pt, pressed state, loading state
`LSChip`, `LSSelect`, `LSTextField`, `LSToggleRow`	DesignSystem	labels from the caller, character counter for prompt
`LSBanner`, `LSToast`, `LSNotesBanner`	DesignSystem	simulation labels use `LSNotesBanner`
`TopBar`, `LSTabBarItems`	CoreUI	
`StoryCard`, `StoryRow`, `StoryArt` (Canvas), `Avatar`	CoreUI	art drawn from `StoryArtSpec`, decorative, hidden from VoiceOver
`LoadingView`	CoreUI	delayed appearance
`NotFoundView`, `EmptyStateView`	CoreUI	
`ParentGated`, `ParentalGateSheet`	CoreUI	§9.3

9.3 Parental gate [A-7.7]



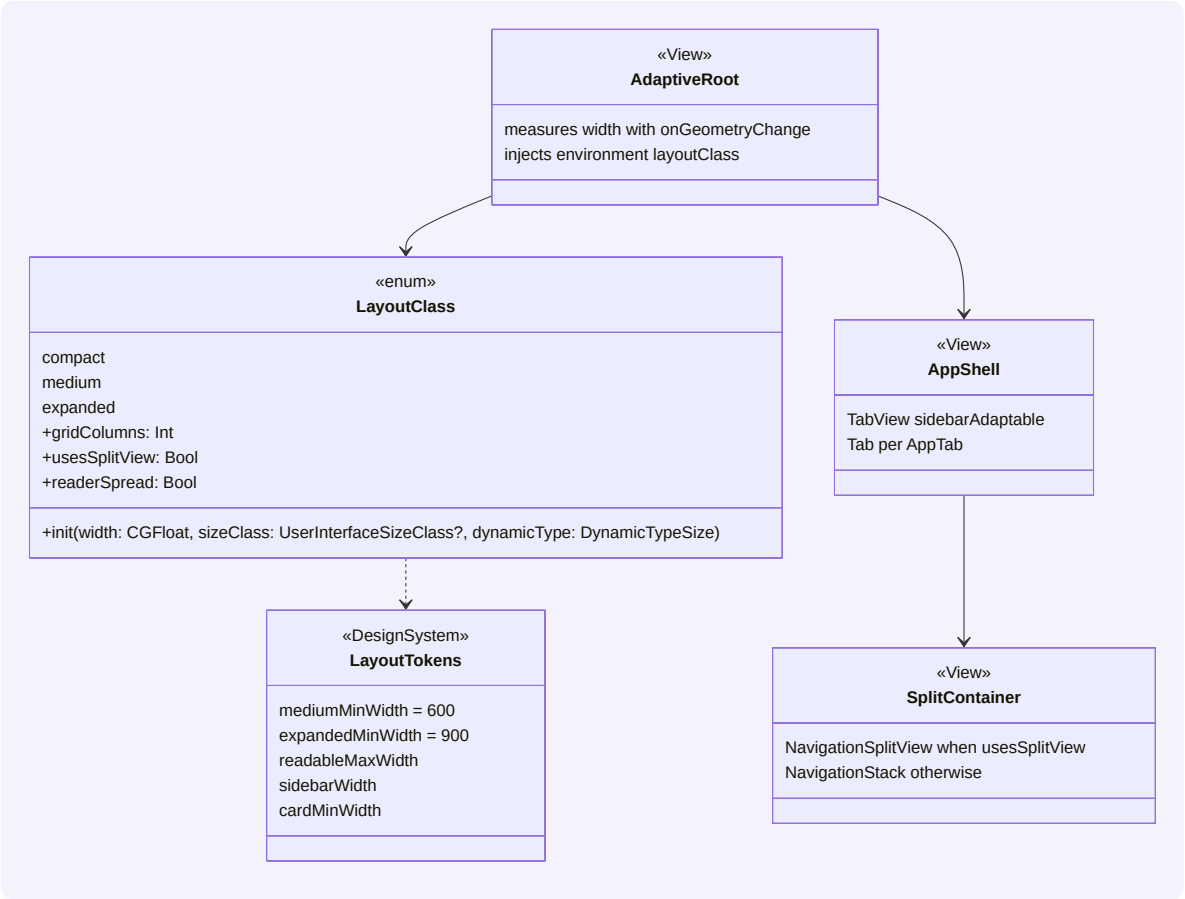
- The keypad is custom (digits only), so a child cannot paste or dictate the answer, and it matches Android.
- Also used before any external link (Help → website) and any future purchase, per App Store guideline 1.3.

9.4 Adaptive layout (iPhone and iPad)

Info.plist and target settings

Setting	Value	Why
`TARGETED_DEVICE_FAMILY`	`1,2`	iPhone and iPad
`UISupportedInterfaceOrientations`	portrait, landscape left, landscape right	iPhone
`UISupportedInterfaceOrientations~ipad`	all four	iPad
`UIRequiresFullScreen`	not set (false)	allows Split View, Slide Over, Stage Manager, resizable windows
`UIApplicationSceneManifest` → `UIApplicationSupportsMultipleScenes`	false in MVP	one window; can be turned on later

Types


«View»
AdaptiveRoot
measures width with onGeometryChange
injects environment layoutClass
«enum»
LayoutClass
compact
medium
expanded
+gridColumns: Int
+usesSplitView: Bool
+readerSpread: Bool
+init(width: CGFloat, sizeClass: UserInterfaceSizeClass?, dynamicType: DynamicTypeSize)
«View»
AppShell
TabView sidebarAdaptable
Tab per AppTab
«DesignSystem»
LayoutTokens
mediumMinWidth = 600
expandedMinWidth = 900
readableMaxWidth
sidebarWidth
cardMinWidth
«View»
SplitContainer
NavigationSplitView when usesSplitView
NavigationStack otherwise

```
        public enum LayoutClass: Sendable, Equatable {
            case compact, medium, expanded

            public init(width: CGFloat, sizeClass: UserInterfaceSizeClass?,
dynamicType: DynamicTypeSize) {
                // Very large text gets more room per item: drop one class so
Kannada at AX sizes never truncates.
                let base: LayoutClass = switch width {
                    case ..<LayoutTokens.mediumMinWidth: .compact
                    case ..<LayoutTokens.expandedMinWidth: .medium
                    default: .expanded
                }
                self = (dynamicType.isAccessibilitySize && base != .compact) ?
base.smaller : base
                if sizeClass == .compact { self = .compact }
            }

            var smaller: LayoutClass { self == .expanded ? .medium : .compact
}

            public var gridColumns: Int { switch self { case .compact: 1; case
.medium: 2; case .expanded: 3 } }
            public var usesSplitView: Bool { self != .compact }
            public var readerSpread: Bool { self == .expanded }
        }

        struct AdaptiveRoot<Content: View>: View {
            @Environment(\.horizontalSizeClass) private var sizeClass
            @Environment(\.dynamicTypeSize) private var typeSize
            @State private var width: CGFloat = 0
            let content: Content

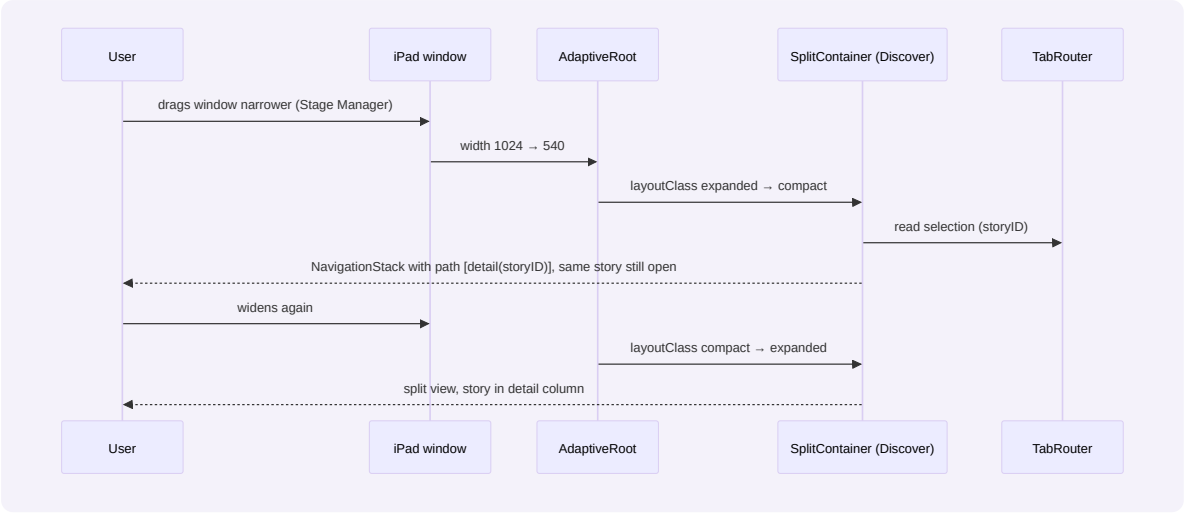
            var body: some View {
                content
                    .environment(\.layoutClass, LayoutClass(width: width,
sizeClass: sizeClass, dynamicType: typeSize))
                    .onGeometryChange(for: CGFloat.self, of: \.size.width) {
width = $0 }
            }
        }

        struct AppShell: View {
            @Bindable var router: AppRouter
            var body: some View {
                TabView(selection: $router.selectedTab) {
                    Tab(value: AppTab.home) { HomeRoot() } label: {
TabLabel(.home) }
                    Tab(value: AppTab.discover) { DiscoverRoot() } label: {
TabLabel(.discover) }
                    // Create, My Stories, Profile …
                }
                .tabViewStyle(.sidebarAdaptable)   // tab bar on iPhone,
sidebar on iPad
            }
        }
```

Rules for every screen

1. Read `@Environment(\.layoutClass)`; never check `UIDevice.current.userInterfaceIdiom` for layout.
2. Grids use `LazyVGrid(columns: [GridItem(.adaptive(minimum: LayoutTokens.cardMinWidth))])` so the column count follows the window.
3. Text blocks are wrapped in `.frame(maxWidth: LayoutTokens.readableMaxWidth)`.
4. Features with list + detail (Discover, My Stories, Settings) use `SplitContainer`; the router keeps one selection, so rotating or resizing keeps the open story.
5. Reader in `.expanded` shows pages `n` and `n + 1`; Next moves by two; progress still stores the left page.
6. Sheets use `.presentationSizing(.form)` on iPad so dialogs (parental gate, confirmations) are not full screen.
7. Keyboard: `.keyboardShortcut` for Next / Previous page (arrow keys), Play / Pause (space), Search (⌘F); menu bar commands on iPadOS 26.
8. Pointer: `.hoverEffect()` on cards and buttons.
9. Every screen ships previews and snapshots for compact and expanded.

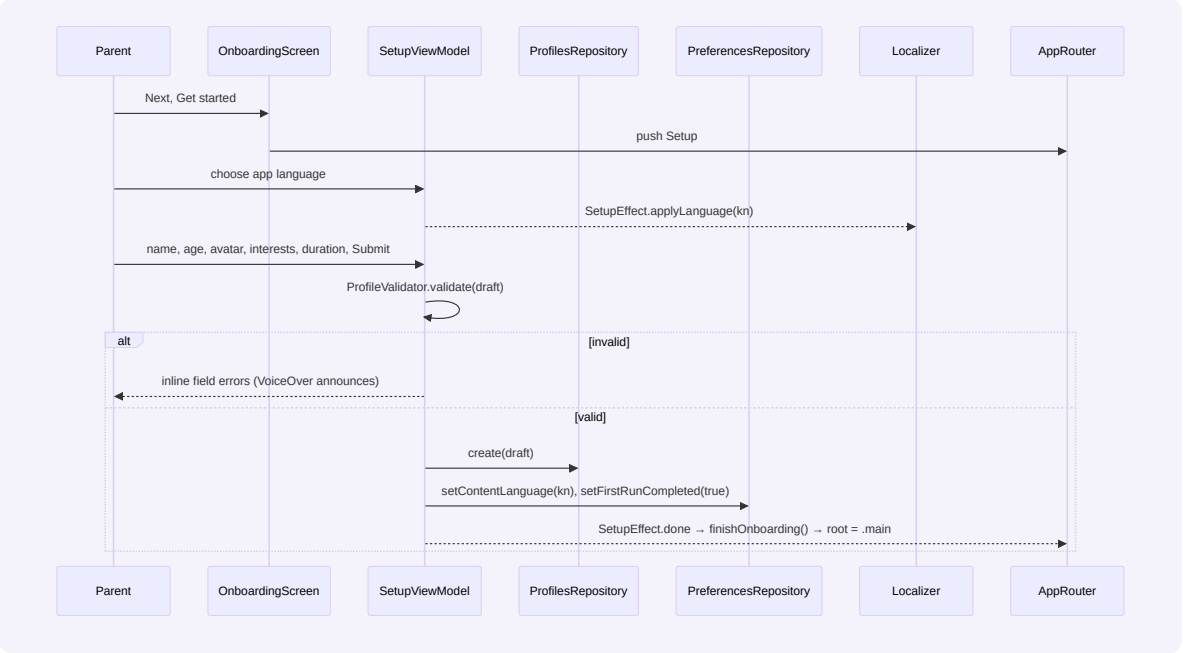
How navigation changes with width



---

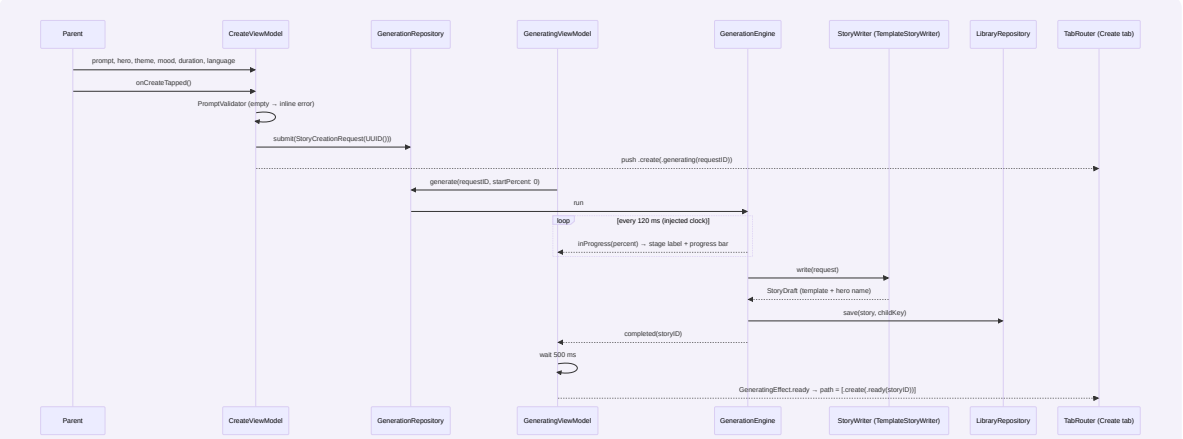
10. Key flows [A-7]

10.1 First run



Skip writes only `firstRunCompleted = true`; the app then browses as the guest.

10.2 Create a story (mocked AI)

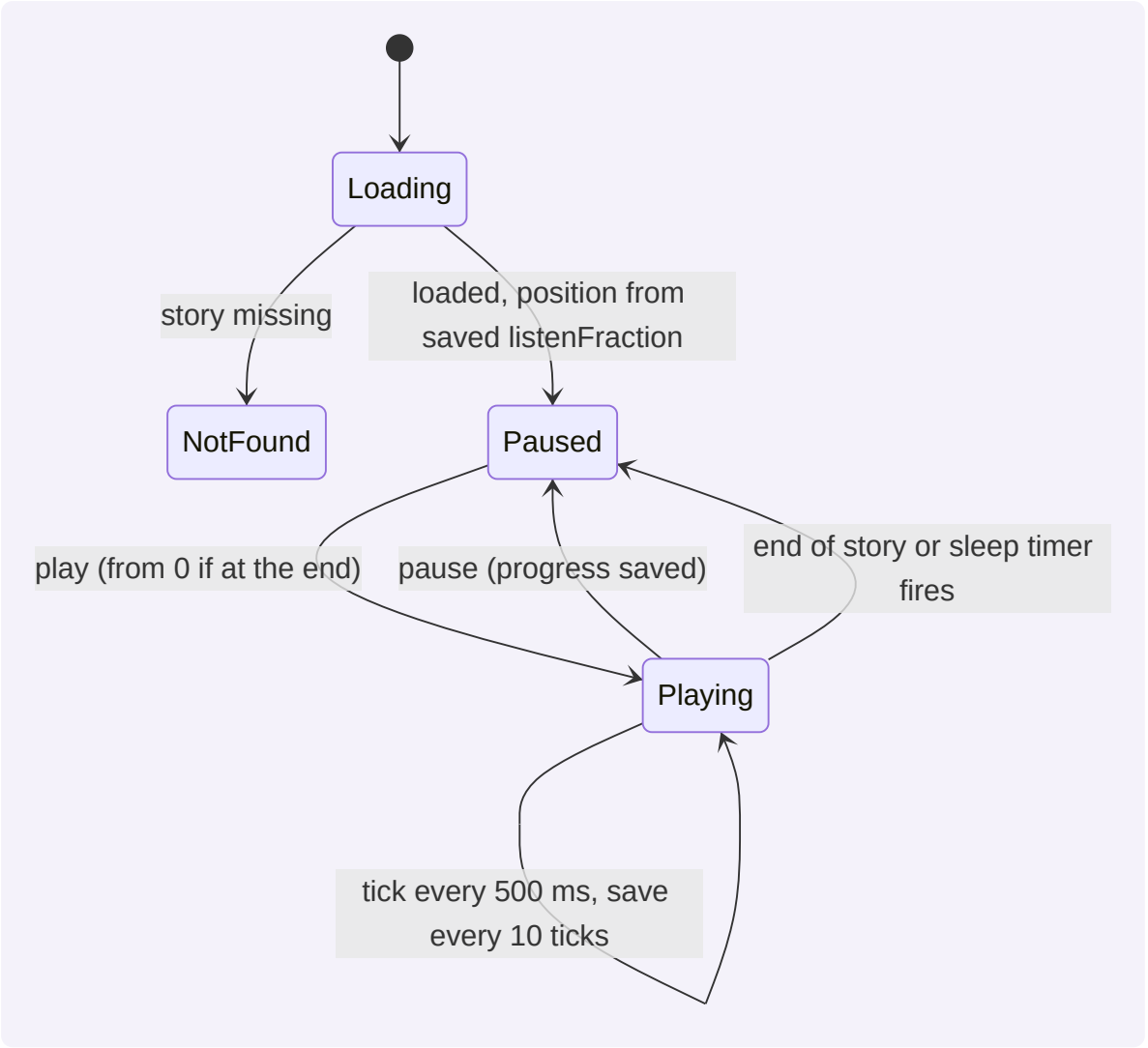


Case	Behaviour
User closes Generating	task cancelled, `paused = true`, **Try again** restarts at 8 %
Back from paused	returns to Create with the prompt intact
App terminated mid-generation	request was in memory only → `.failed(.unknownRequest)` → paused, "Back to your idea"
Writer throws (future real AI)	`.failed(.writer(error))` → paused with retry; error text from catalog

10.3 Reader

- `state = loading | notFound | loaded(ReaderContent)`; `notFound` covers a missing story or one without pages.
- `onNext` / `onPrevious` clamp to the page range and upsert progress, keeping `listenFraction`.
- Text size steps by 0.1 within 0.9...1.3, rounded to tenths, saved in preferences; applied on top of Dynamic Type.
- Play replaces Reader with Player.

10.4 Player



- The tick loop runs in a view-model-owned `Task` using the injected clock.
- `Narrator` events drive position when a real narrator exists; with `SimulatedNarrator` the same events come from the timer, so the view model has one code path.
- The screen keeps the device awake only while playing (`isIdleTimerDisabled`), and restores it on pause or exit.

10.5 Language switch

What	Stored in	Changed by	Effect
UI language	`ui_language_tag` + `Localizer`	Language screen, Setup	every `LText` and formatter re-renders instantly
Story language	`content_language_tag`	Language screen, Setup, Create (per request)	`BrowseContext.languageTag` changes, catalogue streams re-emit

10.6 Delete all local data

Privacy → confirm dialog → `PrivacyViewModel.onConfirmDelete()` → `LocalDataRepository.deleteAll()` → `PrivacyEffect.deleted` → `Localizer.reset()` → `AppRouter.resetToOnboarding()` (all tab paths cleared).

---

11. Error handling [A-8]

Failure	Handled in	User sees
Mock JSON missing or malformed	`ContentResolver` (`try?` per file, logged)	English content; empty list only if the English file is broken (caught by tests)
Missing translated field	`ContentResolver`	that field in English
Unknown story id	`StoryLookup` returns nil	Not Found state with Back
Created story pages unreadable	record mapper	no pages → Not Found
Unknown generation request	`GenerationRepositoryImpl`	paused, "Back to your idea"
Empty or long prompt	`PromptValidator`	inline error, announced to VoiceOver
Invalid profile	`ProfileValidator`	inline field errors
Wrong gate answer	`ParentalGateChallenge`	error and a new question
Store cannot open at launch	`AppContainer`	one retry with a fresh store after moving the old file aside; logged

Errors are typed (`enum XError: Error`) and use Swift 6 typed throws where they cross a protocol. Future network errors follow the Android `error-handling.md` model (`offline`, `timeout`, `server`, `unauthorised`).

---

12. Screen catalogue [A-9]

Screen	Target · route	View model inputs	State · effects
Onboarding (01)	Onboarding · root	—	`OnboardingState(page)` · `finished`
Setup (02)	Onboarding · `.setup(childID?)`	Profiles, Preferences	`SetupState` · `applyLanguage`, `done`
Home (03)	Home · tab root	Catalog, Favourites, BrowseContext, Progress	`HomeState` (resume, featured, categories)
Discover (04)	Discover · tab root	Catalog, Favourites, BrowseContext, SearchStories	`DiscoverState` (query, filter, results)
Category (05)	Discover · `.category(id)`	Catalog, Favourites, Downloads, BrowseContext	`CategoryState` (sort)
Story Detail (06)	Discover · `.detail(storyID)`	StoryLookup, Favourites, Downloads	`StoryDetailState`
Create (07)	Create · tab root	Profiles, Preferences, Generation	`CreateState` · `startGeneration`
Generating (08)	Create · `.generating(requestID)`	Generation	`GeneratingState` · `ready`
Story Ready (09)	Create · `.ready(storyID)`	Library, Favourites, Downloads	`StoryReadyState`
Reader (10)	Reader · `.reader(storyID, page)`	StoryStores, Preferences	`ReaderState`
Player (11)	Reader · `.player(storyID)`	StoryStores, Preferences, Narrator	`PlayerState`
My Stories (12)	Library · tab root	StoryStores	`MyStoriesState(tab)`
Downloads (13)	Library · `.downloads`	Catalog, Library, Downloads	`DownloadsState`
Profile (14)	Settings · tab root	Profiles, Preferences, FeatureFlags	`ProfileState`

Screen	Target · route	View model inputs	State · effects
Subscription (15) 🔒	Settings · `.subscription`	Subscription	`SubscriptionState`
Child profiles (16) 🔒	Settings · `.children`	Profiles	`ChildProfilesState`
Reminder (17)	Settings · `.reminder`	Preferences	`ReminderState` (nothing scheduled)
Notifications (18)	Settings · `.notifications`	Notifications	`NotificationsState`
Launch (19)	Info.plist `UILaunchScreen`	—	—
Language (20) 🔒	Settings · `.language`	Preferences, Localizer	`LanguageState`
Privacy (21) 🔒	Settings · `.privacy`	LocalData, FeatureFlags	`PrivacyState` · `deleted`
Help (22)	Settings · `.help`	AppInfo	—

Each screen has an `XAccessibilityID` enum (`ReaderAccessibilityID.nextPage`) used by UI tests, the iOS twin of Android `XTags` .

---

13. Testing [A-11]

13.1 Seams

Seam	Used by tests to
Repository protocols + `TestSupport` fakes	drive view models with any data
`MockContentSource`	feed real and broken JSON to the resolver
`any Clock<Duration>`	run generation and playback instantly (`TestClock.advance(by:)` from swift-clocks)
`UUID` and `Date` providers	deterministic ids and times
`XScreen(state:actions:)`	snapshot any state without a view model
`AppEnvironment.uiTest`	in-memory store, fixed gate, immediate clock for XCUITest

13.2 Examples

```
        @Test("Generation reaches 100 and saves exactly one story")
        func generationCompletes() async throws {
            let clock = TestClock()
            let library = FakeLibraryRepository()
            let repo = GenerationRepositoryImpl(writer:
TemplateStoryWriter(source: .fixtures), library: library, clock: clock)
            let id = await repo.submit(.fixture(heroName: "Asha"))
            let states = Task { await repo.generate(id: id, startPercent:
0).collect() }
            await clock.advance(by: .seconds(5))
            #expect(await states.value.last == .completed(storyID:
"generated_\(id)"))
            #expect(await library.saved.count == 1)
        }

        @Test("Kannada field missing falls back to English", arguments:
["title", "synopsis", "pages"])
        func fallback(field: String) async { … }
```

13.3 Snapshot matrix

Axis	Values
Language	en, kn
Dynamic Type	large (default), AX5 (Kannada only, iPhone 17 and iPad Air, to keep the count down)
Device	iPhone 17 (compact), iPhone SE 3rd gen (smallest), iPhone 17 landscape, iPad Air 11-inch portrait (medium), iPad Pro 13-inch landscape (expanded), iPad half split (compact on iPad)
Appearance	light (night screens are their own design)

Baselines live in `__Snapshots__/`, are recorded on the CI simulator image only, and are approved by the owner.

13.4 UI journeys (XCUITest)

1. First run in English: onboarding → setup → home shows the child's name.
2. First run in Kannada: same, every visible label is Kannada.
3. Create a story → generating → ready → read two pages → play.
4. Favourite on Detail shows on My Stories.
5. Switch UI and story language without restart.
6. Parental gate blocks Privacy; delete all data returns to onboarding.
7. iPad: sidebar navigation, Discover split view opens Detail in the detail column, Reader shows a two-page spread in landscape, rotating keeps the current page.

---

14. Tooling and CI

14.1 Scripts (`Tools/` , Swift scripts or shell, all free)

Tool	Does
`check-dependencies`	fails on imports outside §1.3
`check-hardcoding`	fails on `Text("…")` literals, `Color(red:…)`, raw numbers in padding/frame, `.font(.system(size:))` outside DesignSystem
`check-translations`	parses every `.xcstrings`: each key has `en` and `kn`, placeholders match, no empty values
`gen-tokens`	`design-tokens.json` → `Tokens.generated.swift`
`gen-l10n`	`.xcstrings` → `L10n` key constants
`import-android-strings`	Android `strings.xml` (values, values-kn) → `.xcstrings` with mapped keys
`sync-android-content`	copies mock JSON from a checkout of the Android repo and diffs it

14.2 `ci.yml` (shape)

```yaml
name: CI
on:
  pull_request: { branches: [development, staging, main] }
  push: { branches: [development, staging, main] }
permissions: { contents: read }
jobs:
  verify:
    runs-on: macos-26              # arm64; free for public repositories
    steps:
      - uses: actions/checkout@v7
      - name: Select Xcode 27.2 (runner default is 26.6)
        run: sudo xcode-select -s "/Applications/Xcode_$(cat .xcode-version).app"   # .xcode-version = 27.2
      - name: Lint tools at pinned versions
        run: Tools/install-lint-tools.sh   # SwiftLint 0.65.1, SwiftFormat 0.64.6 (pinned release binaries)
      - run: swiftformat --lint . && swiftlint --strict
      - run: swift Tools/check-dependencies.swift && swift Tools/check-hardcoding.swift && swift Tools/check-translations.swift
      - run: >
          xcodebuild test -scheme LittleStory
          -destination 'platform=iOS Simulator,name=iPhone 17'
          -destination 'platform=iOS Simulator,name=iPad Air 11-inch (M3)'
          -enableCodeCoverage YES -resultBundlePath TestResults
      - uses: actions/upload-artifact@v7
        if: always()
        with: { name: test-results, path: TestResults.xcresult }
```

Versions above are the newest stable on 11 Nov 2026. Simulator names are checked against `xcrun simctl list` on the runner at kick-off. If the `macos-26` image does not yet carry Xcode 27.2, the job uses the `xcode-27` runner label. No secrets are used.

---

15. Coding conventions

1. Swift API Design Guidelines; types `UpperCamelCase`, members `lowerCamelCase`, acronyms uniform (`storyID`, `URL`).
2. One primary type per file; file name equals type name.
3. `public` only on what other targets need; everything else `internal` or `private`.
4. Value types by default; classes only for `@Observable` view models, routers and `@Model` records.
5. No force unwraps or `try!` outside `AppContainer` launch (lint rule).
6. No singletons (`static let shared`) except `Logger` categories.
7. No `print`; use `Logger` with privacy `.private` for any user data.
8. Every public protocol has a fake in `TestSupport`.
9. Previews use `XScreen` with fixture state in EN and KN.

10. A PR adds tests with the change and passes all gates locally (`make verify` wraps the CI steps).

---

16. Extension recipes

Change	Touch only
New screen	route case + view + view model + state + strings (en, kn) + accessibility IDs + tests + snapshots in the owning feature; one mapping line in the app router
New language	§8.3
New content source (backend)	new `CatalogRepository` implementation in `Data`; one line in `AppContainer`
Real AI writer	new `StoryWriter` implementation behind `FeatureFlags`; parental consent and moderation first (Android `families-compliance` notes)
On-device voice	`SpeechNarrator: Narrator` over `AVSpeechSynthesizer`; check Kannada voice availability per device, fall back to `SimulatedNarrator`
Accounts	new `FeatureAccount` target and `AuthRepository` protocol, following Android PR #20
New setting	key in `PreferencesStore` + property on `PreferencesRepository` + UI in Settings

Appendix A. Android → iOS mapping

Android	iOS
Gradle modules + `build-logic`	Swift package targets + `check-dependencies`
Hilt `@Inject`, `@HiltViewModel`	constructor injection, `AppContainer` factories
`ViewModel` + `StateFlow`	`@Observable @MainActor` class + `state` property
`collectAsStateWithLifecycle`	`.task { await vm.start() }`
`Channel` effects	`AsyncStream<Effect>`
`Flow`, `combine`, `flatMapLatest`	`AsyncStream`, `combineLatest` (AsyncAlgorithms), task restart
`kotlinx-coroutines-test` virtual time	injected `Clock` + `TestClock`
Navigation Compose typed routes	`NavigationStack` + `Hashable` route enums

Android	iOS
Room + DAOs	SwiftData `@Model` + `@ModelActor` (GRDB fallback)
DataStore	`UserDefaults` actor
assets JSON + kotlinx.serialization	bundle resources + `Codable`
`strings.xml` + `values-kn`	`.xcstrings` with `en` and `kn`
`UiText`	`LocalizedStringResource` + `LText`
`AppCompatDelegate.setApplicationLocales`	`Localizer` override + system per-app language page
Compose `Canvas` story art	SwiftUI `Canvas`
Robolectric + Compose UI tests	Swift Testing view model tests + XCUITest
Roborazzi screenshots	swift-snapshot-testing
Detekt, ktlint, Lint, Kover	SwiftLint, SwiftFormat, `xccov` coverage gate
`testTag` / `XTags`	`accessibilityIdentifier` / `XAccessibilityID`

Appendix B. Proposed ADRs for the iOS repo

ADR	Decision
ADR-iOS-001	Native Swift + SwiftUI, not KMP or Compose Multiplatform
ADR-iOS-002	One local package, target per layer and feature, enforced imports
ADR-iOS-003	MVVM with `@Observable`, one state struct, method per action; no TCA
ADR-iOS-004	SwiftData behind repositories; GRDB fallback
ADR-iOS-005	Constructor injection with a composition root; no DI framework
ADR-iOS-006	String Catalogs + `Localizer` in-app language override
ADR-iOS-007	Shared contracts with Android (content JSON, tokens, keys, test IDs)
ADR-iOS-008	Mocked AI behind `StoryWriter`, `Narrator`, `Illustrator`
ADR-iOS-009	Minimum iOS / iPadOS 26; build with Xcode 27.2 and the iOS 27.2 SDK
ADR-iOS-011	One universal iPhone + iPad app with layout classes (compact, medium, expanded), sidebar-adaptable tabs and split views

ADR	Decision
ADR-iOS-010	Swift Testing + swift-snapshot-testing + XCUITest; owner-approved baselines